23 Biophilic Design from a Psychology and Neuroscience Perspective

Restorative Influences on Functional Brain Connectivity

Joel Martínez-Soto, Ryadi Adityavarman, Fernando A. Barrios, and Leopoldo Gonzales-Santos

Biophilic design principle celebrates the fundamental human need to maintain an innate symbiotic relationship with nature within the context of the built environment. Contemporary architectural design tends to overtly focus on the formal composition of the built environment, and in the process, often disregard its essential connection need with nature. This problematic negligence is partly caused by the lack of seamless creative integration between design compositional strategies and biophilic design principles during the typical design process. On the positive side – in tune with contemporary societal and architectural orientation – several designers value the importance of biophilic design and sought to embrace relevant applicable theories from environmental psychology and behavioural sciences to support their design intention on creating restorative-built environment for the holistic well-being of the users. However, other designers tend to make routine design decisions based on primarily creative intuition from a wealth of compositional repertoire gained through extensive design experience in typical design processes.

While this creative-intuitive approach has been quite effective, there is a continuing need for deeper understanding of meanings of design through scientific, evidence-based research on biophilic design. A combination of normative-based design strategies and empirical-based theories provides an optimal integration of biophilic design principles to achieve a creatively designed restorative built environment. Using a combination of a neuroscientific and psychological perspective within environmental behaviour design (EBD) research initiatives, this chapter explains the functional human brain connectivity analysis during the interaction process with natural elements in a restorative built environment.

For most people living in big cities, contemporary urban settings have been perceived and experienced as a source of physical and mental stress. Recent studies show that consistent positive presence of human-nature interactions may influence better adaptive capacity of the human brain for a faster stress recovery process from hectic daily urban circumstances such as noise, traffic, and environmental pollution (Ancora et al., 2022; Jalilisadrabad et al., 2023; Sudimac et al., 2022). People's aspirations for a connection with nature reflect one of the most fundamental human needs. According to the Biophilia hypothesis, this affinity is based on an innate, genetically oriented need (Kellert, 2005; Wilson, 2010). Biophilic design incorporates various natural aspects to create a connection between the designed environments and the natural world (Jung et al., 2023).

Exposure to biophilic designs may provide psychological benefits similar to those obtained by directly interacting with nature. People tend to prefer interior spaces designed with integrated natural elements compared to buildings with a predominant artificial quality (Coburn et al., 2019). Strategies for incorporating natural elements in interior architecture include windows for

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viewing nature (Yin et al., 2020), integration of daylight in addition of artificial lighting (Yin et al., 2018), inserting plants and flowers (Ortegón-Cortázar & Royo-Vela, 2019), providing acoustical experiences of nature-based sounds (Camacho-Guzmán et al., 2023), nature-themed paintings (McSweeney et al., 2021), application of natural materials such as wood (Shen et al., 2020), furniture design with natural finishes and organic design character (e.g., those with curvilinear forms) (Dazkir & Read, 2012) and nature inspired of artwork (Nejati et al., 2016). These have been found to have positive effects on the physiology, emotion, and cognition of the users, suggesting a set of benefits of biophilic design for physical and mental health wellness (Gould Van Praag et al., 2017; Roskams & Haynes, 2020).

Most people spend most of their lives in indoor environments. Indoor lifestyles can contribute to sedentary behaviour, and hence, the development of chronic illness (Beyer et al., 2018). Likewise, mental health can be related to a balance of activities between indoor and natural environments (Catalan et al., 2023). Therefore, innovative designs connecting nature and interior spaces can boost people's physical and mental health. According to Hartig et al. (2008), environmental design can incorporate: (a) protective environmental design (PED), (b) instorative environmental design (IED), and (c) restorative environmental design (RED). PED includes elimination and mitigation of some environmental demands (e.g., exposure to rain and snow), IED strengthens the ability of people to meet subsequent demands (e.g., playgrounds that stimulate learning to develop new capabilities), and RED helps with restoration needs (e.g., through psychological benefits of indoor plants) (Bringslimark et al., 2009).

A plethora of studies on psychological restoration have documented the restorative capacity of biophilic design settings (Martínez-Soto et al., 2021; Nejati et al., 2016; Yin et al., 2020). These studies investigate the influence of environments on the capacity and adaptation of psychological restoration. Psychological restoration is the process of replenishing and restoring psycho-physiological well-being that may be diminished or depleted due to the demands of stressful urban life (Hartig, 2004). A range of restorative experiences may promote positive mood change, renewal of focus attention capacity, and contemplation of the individual self-identity (Korpela et al., 2001). Research in the restorative process investigates attention restoration (Kaplan & Kaplan, 1989) and stress recovery (Ulrich et al., 1991). According to the attention restoration theory, attentional restoration (from mental fatigue) occurs through exposure to calming and fascinating natural stimuli, allowing a tired attentional system to rest. The psycho-physiological stress reduction framework suggests that experiencing natural environments tends to reduce stress by producing positive effects such as sustaining non-vigilant attention, inhibition of negative thoughts, and returning physiological arousal to moderate levels (Ulrich et al., 1991); restorative effects of biophilic design are related to reducing physiological stress (Yin et al., 2020) and anxiety (Yeom et al., 2021). Additionally, the biophilic theory suggests that human brain responses to natural settings reflect more pleasure and relaxation due to an evolutionary affinity for nature (Ancora et al., 2022).

Human Brain Response to Nature and Built Settings

Empirical research shows that the intersection between neuroscience, psychology, and design disciplines allows relevancy analysis on restorative (nature and built) environments (Karakas & Yildiz, 2020). Neuroscience research provides a deeper insight beyond behavioural analyses, explaining brain-neural mechanisms (Farling, 2015) linked to experiences from exposure to built and natural environments.

Brain imaging technology illustrates adaptive human brain neural processes during behavioural interactions in both natural and built environments (Engemann et al., 2019). It is

imperative to identify insights on the influence of restorative natural elements in functional brain connectivity and to highlight its significance in the biophilic design principles.

Functional magnetic resonance imaging (fMRI) can visually record the entire brain networks for a specific activity, enabling the identification of neural centres related to landscape perception (Henderson et al., 2007), visual aesthetic perception (Cela-Conde et al., 2004), and scene preferences (Brielmann et al., 2022).

Table 23.1 illustrates a set of fMRI studies to analyse brain activity in response to natural and built settings. It documents the relevance of environmental and neuroscientific experimental findings on the cognitive, affective, and behavioural outcomes. Spatial design experiences are included based on the theoretical model of architectural appreciation (Coburn et al., 2017). According to this model, architecture engages multiple sensory networks with three systems contributing to perception and appreciation of architecture: (a) sensory-motor systems, (b)

Table 23.1 fMRI studies on human brain response to nature landscapes and built environments.

Source	*Environmental variables*	*Comparative analysis*	*Brain area activation*	*Implications for the perception and appreciation of spatial experiences*	*Behavioural experience related to landscape evaluation*
Martínez-Soto et al. (2013)	Built environments, urban green spaces, natural environment	High restorative potential vs. low restorative potential environments	Left middle frontal gyrus, right middle temporal gyrus, left inferior temporal gyrus, right insula, right inferior parietal lobe, and cuneus	Sensory motor system: visual processing, visuo-spatial exploration, emotion-valuation system	Novelty-seeking behaviour, storing of spatial information, focusing attention, visual perception, awareness of emotionally charged stimuli, approach-related behaviours
Kim and Jeong (2014)	Comfortable residential sceneries: pastoral locales, gardens, and forests. Uncomfortable residential scenery: buildings in the background, a smoggy sky, and busy, traffic-jammed streets with a dense population.	Comfortable residential sceneries vs. uncomfortable residential scenery	Calcarine gyrus	Sensory motor system: visual processing, emotion-valuation system	Emotion excitation and visual information processing
Tang et al. (2017)	Pictures of Mountain, flowers, and water	Urban vs. water or mountain landscapes pictures	Cuneus, right cingulate gyrus and left precuneus	Sensory motor system: visuo-spatial exploration	Visual processing
Seiyama et al. (2018)	Spatial, colour, shape, and depth information from landscape pictures	Pleasant landscape pictures vs. unpleasant unattractive landscapes	Ventral pathway and left lingual gyrus. Dorsal visual pathway, right precuneus for traditional architecture	Sensory motor system: sensory processing, emotion-valuation system towards the environment	Spatial processing, cognitive appraisal, unpleasant emotion generation

(*Continued*)

Table 23.1 (Continued)

Source	*Environmental variables*	*Comparative analysis*	*Brain area activation*	*Implications for the perception and appreciation of spatial experiences*	*Behavioural experience related to landscape evaluation*
Chang et al. (2021)	Street green cover density (trees, bushes, grass)	Intact pictures (low-, medium-, high-density cover vegetation) vs. scrambled pictures	Superior parietal lobule, superior temporal gyrus, middle frontal gyrus, cuneus	Sensory motor system: sensory processing and cognitive processing	Spatial processing, executive attention, and sensory encoding
Zhang et al. (2019)	Pictures of landscape gardens and natural landscapes	Pictures of landscape gardens vs. natural landscapes	Inferior occipital lobe, left superior parietal lobule, right fusiform gyrus, right cuneus, and right hippocampus	Sensory motor system: visuo-spatial exploration	Processing of the configuration of shapes and objects, visuo-spatial coding.
Sudimac et al. (2022)	Urban forest and a busy street with shopping malls	60-min walk in natural vs. 60-min walk in urban environment	Amygdala	Emotion-valuation system	Amygdala activity decreased after the walk-in nature, but only in women

Source: Joel Martínez-Soto

knowledge-meanings systems, and (c) emotion-valuation systems. The sensory-motor systems include the sensory networks – that process perceptual features (e.g., visual and nonvisual architecture experiences). It also incorporates a set of motor responses related to sensory-spatial explorations and reward-based behaviours to approach or avoid a space. Knowledge-meanings systems refer to the role of education and memories in the cognitive and affective responses towards a built environment (expertise levels, person's past experiences, expectations, etc.). Finally, emotion valuation systems incorporate evaluative and emotional responses to the spatial experiences.

According to the neuroscientific evidence in Table 23.1, the mere exposure to natural environments elicited brain activation in the areas related to visual processing, spatial processing, cognitive appraisal, sensorial integration, emotion generation, and motivational behaviour. Compared to non-natural city-built artefacts, most experiences of nature (e.g., viewing scenes with trees) can sustain a better attentional functioning thanks to the cognitive mechanisms related to attention and reflection (Jiang et al., 2023; Martínez-Soto et al., 2013). Similarly, exposure to natural features through single walks can have salutogenic effects on stress and related brain regions (Sudimac et al., 2022).

Restorative influences on resting-state brain networks

Overall, functional brain connectivity analyses enhance a deeper understanding of the functional interaction among various brain regions compared to localised brain activation (Adhikari, 2014). Studies on functional brain connectivity analyse the temporal activity between several brain networks (e.g., the denominated resting state brain networks) during a cognitive, emotional, pharmacological condition (stimulus affecting the central nervous system because of medicine) or in a rest condition (task independent activity). Brain connectivity resting state networks (Andrews-Hanna et al., 2010) are shown in Figure 23.1.

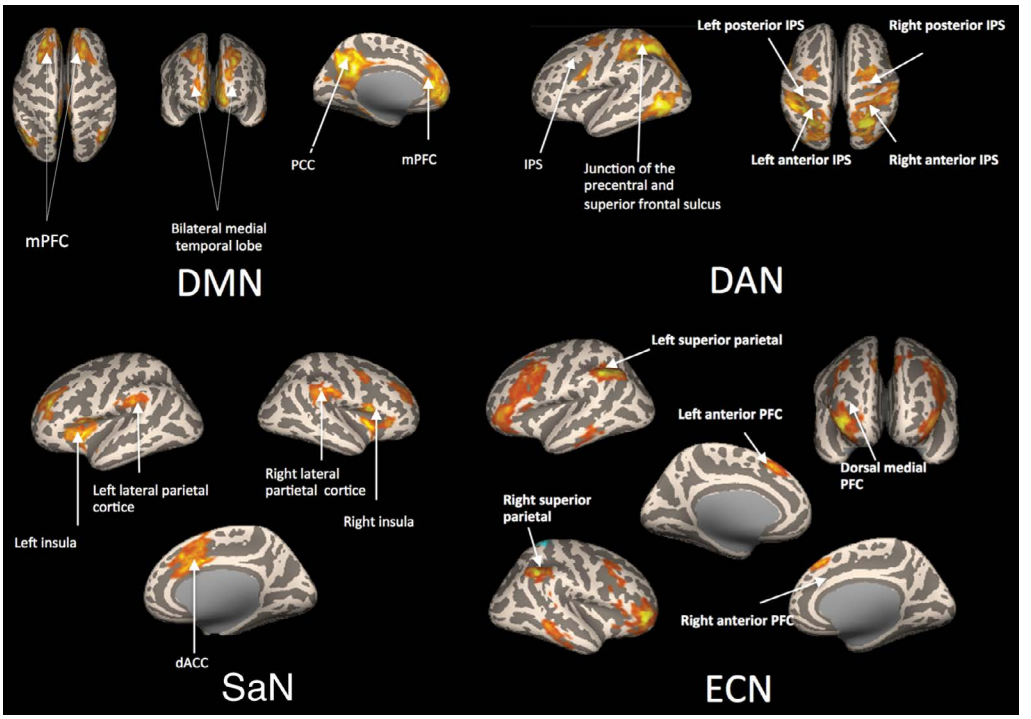


Figure 23.1 The brain connectivity resting state networks. DMN, default mode network; DAN, dorsal attention network; SaN, salience network; ECN, executive control network. mPFC, medial prefrontal cortex; PCC, posterior cingulate cortex; IPS, intraparietal sulcus; dACC, dorsal anterior cingulate cortex; PFC, prefrontal cortex.

Source: Fernando Barrios

Research on the neural correlates of exposure to restorative environments investigates environmental factors influence on brain's recovery capacity from stress and cognitive well-being. Brain connectivity on the resting state networks after exposure to restorative environments has been explored regarding the functional connectivity characteristics between the components of the resting state networks. Based on Figure 23.1, Table 23.2 represents a brief synthesis of brain functional connectivity studies on restorative settings using fMRI resting state analysis of connectivity.

In contrast to the outcomes of Table 23.1, which considers brain regional activity, Table 23.2 shows a deeper analysis of the spontaneous neural activity of functional interactions between brain regions involved with restorative experiences, thus supporting the empirical and theoretical basis of the attentional restoration and stress recovery theories. Most of the settings contain environmental features involving environmental features (water, plants, views and vistas, habitats, and urban ecosystems), leaving aside the incorporation of other elements of biophilic design not covered in the previous literature. However, the previous findings are relevant to EBD and biologically based design into forms of restorative environmental design considering thus the beneficial contact between people and nature-environmental features in landscapes and modern buildings (Kellert et al., 2008).

Table 23.2 Summary table on the restorative influences on brain connectivity according to the literature

Resting state network	*Anatomy*	*Function*	*Source*	*Environmental condition*	*Comparative analysis*	*Brain connectivity outcomes*	*Spatial experience*
Default Mode Network (DMN)	Posterior cingulate cortex, medial prefrontal cortex, left and right lateral parietal cortices, bilateral medial temporal lobe regions	Episodic memory recall, future thinking, social tasks, self-related processing, and mind wandering	Hunter et al. (2010)	Naturalistic tranquil condition (e.g., beach images) and non-tranquil condition (freeway audiovisual images)	Tranquillity associated vs. nontranquillity associated	Engagement of the resting state network by sensory systems. Connectivity of the DMN components: medial prefrontal cortex, posterior cingulate gyrus	Self-reflection, subjective experiences of tranquillity, affective dimensions of auditory perception. Monitoring and visual working memory during naturalistic exposure
			Gould Van Praag et al. (2017)	Soundscapes of artificial and natural environments	Naturalistic vs. artificial sounds vs. no soundscape condition control	Increase middle insula activation. Increase connectivity between posterior cingulate cortex and precuneus and decrease in connectivity between the PCC and the medial prefrontal cortex	A reduced capacity for external attentional monitoring and an increase in attentional capture by artificial soundscape. Attenuation of self-referential thought processes during exposure to naturalistic environmental stimuli
			Kühn et al. (2021)	Natural and built environment images	Natural vs. built environment settings	Functional network connectivity at rest was higher when participants were seeing natural vs. built environmental pictures in the nodes corresponding to the networks: DAN and DMN and DMN and somatomotor	Oriented cognitive processing

(*Continued*)

Table 23.2 (Continued)

Resting state network	*Anatomy*	*Function*	*Source*	*Environmental condition*	*Comparative analysis*	*Brain connectivity outcomes*	*Spatial experience*
Salience Network (SaN)	Anterior insula and dorsal anterior cingulate cortex. Includes three key subcortical structures: the amygdala, the ventral striatum, and the substantia nigra/ventral tegmental area	Plays a fundamental role in awareness, tonic alertness, response towards painful stimuli and empathy for pain and metabolic stress. Intrinsic functional connectivity in the salience network correlated with anxiety	Jiang et al. (2023)	Videos of green infrastructure (trees and bioswales)	Urban sites without green infrastructure vs. urban sites with trees	Higher functional connectivity between the anterior cingulate cortex and the right middle occipital gyrus	Increased vigilance, reduced attentional functioning
			Gould Van Praag et al. (2017)	Soundscapes of artificial and natural environments	Naturalistics vs. artificial soundscapes	Increased connectivity with limbic structures in the naturalistic condition	Increase in emotional salience under natural conditions
Dorsal Attention Network (DAN)	The dorsal system is bilateral and consists of the intraparietal sulcus and the junction of the precentral and superior frontal sulcus (frontal eye field) in each hemisphere	It is involved in voluntary (endogenous, top-down) orienting attention.	Kühn et al. (2021)	Natural and built environment images	Natural vs. built environment settings	Functional network connectivity at rest was higher when participants were seeing natural vs. built environmental pictures in the DAN and ventral attention network	Oriented cognitive processing
Fronto-Parietal Control Network	Lateral prefrontal cortex (LPFC) and posterior parietal cortex	Attentional control and regulation of cognitive processes. It can inhibit the DMN when it is relevant to task performance	Jiang et al. (2023)	Videos of green infrastructure (trees and bioswales)	Urban sites with trees vs. urban sites without trees	Increased functional connectivity in the right lateral prefrontal cortex and the occipital cortex	Best attentional functioning. Top-down attention and cognitive processes

Source: Joel Martínez-Soto

Discussion

Interior architectural design has an important effect on brain neurological responses and patterns of human behaviour (Eberhard, 2009), emphasising the need to consider biologically based, biophilic spatial design (Coburn et al., 2017). The present study indicates neurobiological principles from the interaction in the context of natural and built environment. Empirical research contributes towards the understanding of the environmental/protective factors on potential recovery adaptive response promoting wellness. Additional systematic and focused research on human behavioural response in built environments is needed to better understand well-designed spaces promoting harmonious human-environment relationships (Karakas & Yildiz, 2020).

As referred in the previous studies, understanding aesthetic experiences requires examining the biological basis of human needs to identify different outcomes from salutogenic design of spaces (Brielmann et al., 2022). Neuroscientific research on neural mechanisms linked to mental restoration and stress recovery provides clear guidance for architectural designers to increase interior biophilic spaces (Ancora et al., 2022). Applied research from psychological restoration theory and biophilic design principles ensure mitigate the negative effects of interior built environments on the quality of human health and wellness. As previously pointed out, the influence of restorative natural elements in functional brain connectivity highlights its significance in biophilic design principles and design interventions. The incorporation of physical elements, patterns, and materials that resemble people's connectivity with nature into biophilic design promotes the integration of healthy interior spaces and the configuration of nature-design relationships in interior built environments (Ryan et al., 2014). In essence, the integration of biophilic attributes within interior built spaces has the potential to enhance sensory, auditory, and visual experiences. These attributes can include natural elements (e.g., daylight) and artificial qualities (e.g., artificial lighting). The incorporation of biophilic attributes can facilitate a more profound connection with the designed environment, the natural world, and human needs.

Table A Glossary of terms

Key Concept	
Functional magnetic resonance imaging (fMRI)	A technique for measure and mapping hemodynamic responses after enhanced neural activity
Functional brain connectivity	Represents how spatially separated brain regions correlate each other over time
Task dependent activity	Involves measures of brain activity during the performance of a given task
Task independent activity	Task free (spontaneous) measure of brain activity
Brain resting state activity	Intrinsic brain activity. It represents a brain's base line activity in the absence of goal-directed and external stimuli
Resting state networks (RSNs)	Functionally related sub networks during rest
Resting state functional connectivity (rsFC)	An approach that helps to researchers to study the brain functional organization

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